

AI INTERVIEW COACH

Real-Time Mock Interview Analysis & Feedback System

"Practice with Data. Speak with Confidence."

What Is This Project?

Most students never get proper interview practice before facing real recruiters. A friend giving feedback is biased, and professional mock interviews are expensive or hard to access. This app fixes that: a student records a practice interview on their phone, and the app instantly analyzes how they spoke and how they looked — giving an honest, data-based report so they can improve before the real thing.

In short: **record your practice answer** → **get instant, honest feedback** → **improve every time**.

Why It Matters

- Placements decide careers — yet most students walk in with zero real practice.
- Nervous habits (filler words, low eye contact, fast speech) go unnoticed without feedback.
- Available anytime, free, and judgment-free — unlike human mock interviews.
- Turns a soft skill (communication) into something measurable and improvable, like a fitness tracker for interviews.

How It Works — Step by Step

1. Pick a Question	Student chooses a question from the built-in bank — HR, technical, or role-specific.
2. Record the Answer	Using the phone camera, the student records their spoken answer, just like a real interview.
3. Speech Is Analyzed	The app converts speech to text and checks pace, filler words ("um", "like"), and tone.
4. Face & Body Language Checked	A vision model checks eye contact and expression during the recording.
5. Answer Quality Scored	The answer is compared against an ideal structure (e.g. STAR method for behavioral questions).
6. Feedback Report Generated	A simple report shows scores, specific tips, and charts — e.g. "Said 'um' 14 times, eye contact 40%".
7. Progress Tracked Over Time	Every session is saved so the student can see themselves improve across attempts.

Project Roadmap

A realistic timeline for a semester-long mini project (adjust to your actual ISE-1 / ISE-2 dates):

Phase	Duration	Deliverable
Phase 1 — Research & Design (maps to ISE-1)	Week 1-4	Problem statement, SRS document, ER diagram, class diagram, wireframes, question bank drafted
Phase 2 — Core App Build	Week 5-7	Android UI, camera recording module, question bank integration, local storage setup
Phase 3 — Speech Analysis	Week 8-9	Speech-to-text integration, filler word detection, pace & tone scoring
Phase 4 — Face/Body Analysis	Week 10-11	Eye contact & expression detection using on-device ML Kit
Phase 5 — Scoring & Report Dashboard	Week 12-13	Answer-quality scoring, combined report screen, progress charts
Phase 6 — Testing & Polish (maps to ISE-2)	Week 14-15	Bug fixes, UI polish, full working demo, final project report & presentation

Simple Tech Stack Overview

Layer	Tool/Tech	Purpose
Mobile App	Android (Kotlin/Java), CameraX	Record video, show reports & charts
Speech Analysis	Speech-to-Text API / Whisper-tiny	Convert speech to text, detect fillers & pace
Face Analysis	ML Kit Face Detection (free, on-device)	Detect eye contact & expression
Backend	Firebase / Node.js	Store sessions, handle API calls
Database	Firebase Firestore / SQLite	Save question bank & session history
Charts	MPAndroidChart	Show progress visually

How It Fits Your Course Outcomes

- **CO1 (Analyze):** Identifying the real problem — lack of accessible interview practice.
- **CO2 (Design):** Designing a scalable app + backend architecture with clear ER and class diagrams.
- **CO3 (Create):** Building the Android UI/UX using CameraX, RecyclerView, and chart libraries.
- **CO4 (Integrate & Deploy):** Connecting speech/ML APIs, testing, and deploying a working demo.